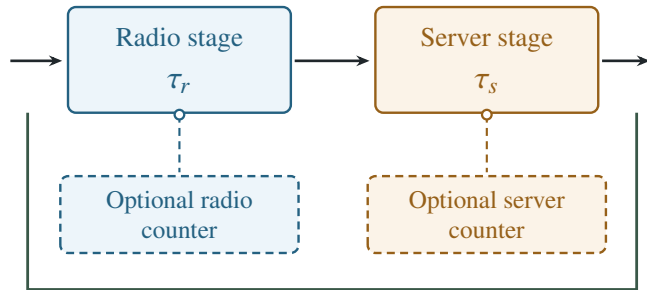


**(a) One radio-server configuration  $x$**

$\theta = (\tau_r, \tau_s)^\top$ : full conditional stage means



Aggregate diagnostic:  $\tau_r + \tau_s$

**(b) What each diagnostic measures**

| Probe type $e$ | Known $a_e$   | Mean $a_e^\top \theta$ |
|----------------|---------------|------------------------|
| Aggregate      | $(1, 1)^\top$ | $\tau_r + \tau_s$      |
| Radio counter  | $(1, 0)^\top$ | $\tau_r$               |
| Server counter | $(0, 1)^\top$ | $\tau_s$               |

$$Y_{e,i} = a_e^\top \theta + \varepsilon_{e,i} \quad \text{mean + report error}$$

The goal specifies required combinations; the instruments specify observed combinations.